Executive Summary: Credit Default Scorecard

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The problem

22.1% of customers default the following month. If one default wipes out the profit from four good accounts (an assumption), the break-even default rate is **20%**, so approving everyone **loses money**.

The finding

A bank-style scorecard built from six variables (repayment status, months late, worst delay, credit limit, utilisation, payment ratio) ranks customers by risk. On held-out data it reaches **AUC 0.75, Gini 0.51, KS 0.40**. Approving only the safest 70% (score of about 523 or more) cuts the default rate among approved customers from 22.1% to **12.7%** and turns a loss into a profit.

Held-out test set (9,000 customers)	Approved	Default rate of approved	Profit (units)
Approve everyone	9,000	22.1%	-955
Approve safest 70% (cut-off set on training data)	**6,361**	**12.7%**	**+2,326**

Loss-to-gain ratio	Break-even default rate	Best approval rate	Profit at cut-off	Profit, approve all
2 to 1	33.3%	80%	+4,135	+3,027
3 to 1	25.0%	70%	+3,133	+1,036
4 to 1	**20.0%**	**70%**	**+2,326**	**-955**
6 to 1	14.3%	50%	+1,060	-4,937

The costlier a default is, the tighter the cut-off should be.

What drives default

- **Late payment dominates.** Customers with any late payment in six months default at 42.7%, against 11.7% for those never late (gap 31 points, 95% CI 30 to 32). Two months late: 69%.
- **Credit limit:** default falls from 36% (limit under 50k) to 15% (200k+).
- **Sex, marital status and education are not used**, and age was dropped as weak. Approval rates at the cut-off are similar for men (69.7%) and women (71.2%).

Recommendation

Use the scorecard to approve roughly the safest 70% of applicants under a 4-to-1 loss ratio, and reset the cut-off once real revenue and loss figures are known.

What this analysis cannot tell you

- **The 4-to-1 loss ratio is an assumption.** There are no real revenue or loss figures.
- **Random split, not out-of-time.** Stability over later periods is untested, and the data is a single 2005 snapshot.
- **Calibration is imperfect:** the safest decile is over-predicted (8.2% vs 5.0% actual), mid deciles under-predicted. Fine for ranking, not for provisioning.
- **The credit limit may reflect the bank's earlier risk view**, so part of its signal may be circular.

Next step

Validate on a later period, add a stability (PSI) check, replace the assumed ratio with real loss data, and run a formal fairness review before any use.

